StudySapiens Board MCQs

Chapter 11 — Human Circulatory System

Board-style MCQs · Questions first, answer key with brief reasons at the end · PECTAA 2025–26

Choose the best option. Each question carries one mark. The answer key with brief reasons is given at the end of this chapter.

1. The human heart is located:

(a) In the abdominal cavity
(b) Between the lungs in the chest cavity
(c) Behind the stomach only
(d) In the neck

2. The sac enclosing the heart is called the:

(a) Meninges
(b) Pericardium
(c) Pleura
(d) Peritoneum

3. The outer fibrous layer of the pericardium:

(a) Contracts to pump blood
(b) Stores bile
(c) Produces blood cells
(d) Protects against shock and prevents over-dilation

4. The pericardial cavity contains:

(a) Pericardial fluid
(b) Lymph only
(c) Air only
(d) Blood only

5. The thick muscular layer of the heart wall is the:

(a) Pericardium only
(b) Endocardium
(c) Myocardium
(d) Epicardium only

6. The inner lining of heart chambers is the:

(a) Epicardium
(b) Endocardium
(c) Myocardium
(d) Pleura

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7. The heart has how many chambers?

(a) Two
(b) Four
(c) Five
(d) Three

8. The valve between the right atrium and right ventricle is the:

(a) Tricuspid valve
(b) Aortic valve
(c) Bicuspid valve
(d) Pulmonary valve

9. The valve between the left atrium and left ventricle is the:

(a) Tricuspid valve
(b) Bicuspid/mitral valve
(c) Pulmonary valve
(d) Aortic valve

10. Semilunar valves are located at the base of the:

(a) Coronary sinus only
(b) Pulmonary artery and aorta
(c) Pulmonary veins only
(d) Venae cavae

11. The left ventricle has a thicker wall because it:

(a) Pumps blood to the whole body
(b) Pumps blood only to lungs
(c) Stores blood only
(d) Receives blood only

12. Deoxygenated blood enters the right atrium through the:

(a) Pulmonary veins
(b) Aorta
(c) Venae cavae
(d) Pulmonary arteries

13. Oxygenated blood enters the left atrium through the:

(a) Pulmonary arteries
(b) Venae cavae
(c) Pulmonary veins
(d) Aorta

14. The pulmonary trunk carries:

(a) Deoxygenated blood to the body
(b) Deoxygenated blood to the lungs
(c) Oxygenated blood to the body
(d) Oxygenated blood to the lungs

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15. The aorta carries:

(a) Deoxygenated blood to the lungs
(b) Deoxygenated blood to the body
(c) Lymph only
(d) Oxygenated blood to the body

16. The relaxed phase of the cardiac cycle is called:

(a) Atrial systole only
(b) Diastole
(c) Systole
(d) Ventricular systole only

17. Atrial systole lasts about:

(a) 1 second
(b) 0.1 second
(c) 0.4 second
(d) 0.3 second

18. Ventricular systole lasts about:

(a) 1 second
(b) 0.3 second
(c) 0.4 second
(d) 0.1 second

19. The "lubb" heart sound is caused by closure of:

(a) Pulmonary valve only
(b) Aortic valve only
(c) Atrioventricular valves
(d) Semilunar valves

20. The "dubb" heart sound is caused by closure of:

(a) Atrioventricular valves
(b) Tricuspid only
(c) Mitral only
(d) Semilunar valves

21. A heart murmur is usually caused by:

(a) Low blood glucose only
(b) Increased lymph flow
(c) Turbulent blood flow through defective valves
(d) Normal valve closure

22. The pacemaker of the heart is the:

(a) Bundle of His
(b) Purkinje fibres
(c) SA node
(d) AV node

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23. The SA node is located in the:

(a) Left ventricle
(b) Right atrium
(c) Right ventricle
(d) Left atrium

24. The AV node is located near the:

(a) Interatrial septum
(b) Aortic valve
(c) Pulmonary valve
(d) Ventricular apex only

25. The delay at the AV node allows:

(a) The heart to stop
(b) Ventricles to contract first
(c) Blood to flow backward
(d) Atria to complete contraction before ventricles contract

26. The bundle of His conducts impulses to the:

(a) Atria only
(b) Ventricles via Purkinje fibres
(c) Liver only
(d) Lungs only

27. An artificial pacemaker is used when the:

(a) SA node is weak or impulses are blocked
(b) Lungs are healthy
(c) Blood pressure is normal
(d) Liver fails only

28. An ECG records:

(a) Electrical activity of the heart
(b) Blood pressure only
(c) Heart sounds only
(d) Oxygen saturation only

29. Arteries carry blood:

(a) Only from lungs
(b) Away from the heart
(c) Toward the heart
(d) Only to lungs

30. Veins carry blood:

(a) Toward the heart
(b) Only to lungs
(c) Away from the heart
(d) Only from lungs

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31. The thickest layer in arteries is the:

(a) Tunica intima
(b) Tunica externa only
(c) Tunica media
(d) Lumen

32. Capillaries are composed of:

(a) Smooth muscle only
(b) Three thick layers
(c) Cartilage
(d) A single layer of endothelial cells

33. The exchange of materials between blood and tissues occurs in:

(a) Aorta only
(b) Arteries only
(c) Venae cavae only
(d) Capillaries

34. Fluid that leaks from capillaries into tissues is called:

(a) Lymph only
(b) Interstitial fluid
(c) Plasma only
(d) Serum only

35. Veins contain valves to:

(a) Filter lymph
(b) Produce blood cells
(c) Prevent backflow of blood
(d) Increase blood pressure

36. All arteries carry oxygenated blood except the:

(a) Coronary arteries
(b) Carotid arteries
(c) Pulmonary arteries
(d) Aorta

37. All veins carry deoxygenated blood except the:

(a) Jugular veins
(b) Hepatic veins
(c) Venae cavae
(d) Pulmonary veins

38. Blood velocity is highest in the:

(a) Veins only
(b) Aorta
(c) Venules
(d) Capillaries

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39. Blood velocity is lowest in the:

(a) Aorta
(b) Vena cava
(c) Arteries
(d) Capillaries

40. Pre-capillary sphincters regulate:

(a) Heart rate only
(b) Breathing rate only
(c) Lymph formation only
(d) Blood flow into capillaries

41. Vasodilation of arterioles:

(a) Forms lymph only
(b) Decreases blood flow
(c) Stops circulation
(d) Increases blood flow to tissues

42. Vasoconstriction of arterioles:

(a) Produces oxygen
(b) Increases blood flow
(c) Decreases blood flow
(d) Has no effect

43. Pulmonary circulation carries blood between the heart and:

(a) Liver only
(b) Kidneys only
(c) Brain only
(d) Lungs

44. Systemic circulation carries blood between the heart and:

(a) Kidneys only
(b) Liver only
(c) All body parts except lungs
(d) Lungs only

45. Coronary circulation supplies the:

(a) Liver
(b) Kidneys
(c) Lungs
(d) Heart wall

46. The hepatic portal vein carries blood from the intestines to the:

(a) Heart directly
(b) Kidneys directly
(c) Lungs directly
(d) Liver

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47. Blood pressure is highest in the:

(a) Vena cava
(b) Venules
(c) Capillaries
(d) Aorta

48. Systolic pressure in a normal young adult is about:

(a) 120 mm Hg
(b) 80 mm Hg
(c) 140 mm Hg
(d) 60 mm Hg

49. Conventional diastolic pressure is about:

(a) 100 mm Hg
(b) 80 mm Hg
(c) 120 mm Hg
(d) 60 mm Hg

50. Blood pressure is measured with a:

(a) Stethoscope only
(b) ECG machine only
(c) Sphygmomanometer
(d) Spirometer

51. Baroreceptors are located in the:

(a) Carotid arteries and aortic arch
(b) Veins only
(c) Capillaries only
(d) Lymph nodes only

52. When blood pressure falls, baroreceptors trigger:

(a) No response
(b) Increased heart rate and vasoconstriction
(c) Vasodilation only
(d) Decreased heart rate only

53. ADH raises blood pressure by:

(a) Increasing urine output only
(b) Decreasing water retention
(c) Lowering heart rate only
(d) Increasing water retention and vasoconstriction

54. Atrial natriuretic hormone (ANH) lowers blood pressure by:

(a) Increasing heart rate only
(b) Constricting arterioles only
(c) Increasing salt and water excretion
(d) Retaining salt and water

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55. Atherosclerosis involves deposition of:

(a) Fatty materials/cholesterol in arteries
(b) Calcium in bones only
(c) Uric acid in joints only
(d) Bile in gallbladder only

56. Arteriosclerosis refers to:

(a) Hardening of arterial walls
(b) Lymph node swelling
(c) Inflammation of veins only
(d) Heart valve infection only

57. A thrombus is:

(a) A heart valve
(b) A plasma protein only
(c) A blood clot within a vessel
(d) A lymph node

58. An embolus is:

(a) A stationary clot only
(b) A platelet only
(c) A white blood cell
(d) A dislodged thrombus that travels in blood

59. Myocardial infarction is:

(a) High blood pressure only
(b) A valve murmur only
(c) Death of heart muscle due to blocked blood supply
(d) Inflammation of pericardium only

60. Angina pectoris is:

(a) Heart muscle death always
(b) Chest pain due to insufficient oxygen to heart muscle
(c) Kidney failure
(d) Lung infection

61. Coronary bypass surgery:

(a) Uses a graft to bypass a blocked coronary artery
(b) Replaces the heart valve only
(c) Opens arteries with a balloon only
(d) Removes the heart

62. Angioplasty uses a balloon to:

(a) Drain lymph
(b) Dilate a narrowed artery
(c) Remove the heart
(d) Replace valves

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63. A stent is:

(a) A mesh tube that keeps an artery open
(b) A heart valve
(c) A lymph vessel
(d) A pacemaker

64. Hypertension is generally defined as blood pressure consistently above:

(a) 120/80 mm Hg
(b) 140/90 mm Hg
(c) 100/60 mm Hg
(d) 90/60 mm Hg

65. Hypertension can damage the:

(a) Lungs only
(b) Liver only
(c) Brain, heart and kidneys
(d) Stomach only

66. The lymphatic system returns excess _____ to the blood.

(a) Interstitial fluid
(b) Air only
(c) Bile only
(d) Urine only

67. Lymph capillaries in intestinal villi are called:

(a) Lacteals
(b) Venules
(c) Arterioles
(d) Sinusoids only

68. Lymph nodes filter:

(a) Blood only
(b) Urine only
(c) Lymph
(d) Bile only

69. The spleen filters:

(a) Urine only
(b) Blood
(c) Bile only
(d) Lymph only

70. The thoracic duct empties lymph into the:

(a) Left subclavian vein
(b) Aorta
(c) Right subclavian vein only
(d) Pulmonary vein

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71. The right lymphatic duct empties into the:

(a) Right subclavian vein
(b) Left subclavian vein
(c) Inferior vena cava only
(d) Aorta

72. Lymph flow is helped by:

(a) Heart pumping directly
(b) High arterial pressure only
(c) Skeletal muscle contractions and valves
(d) Gravity only

73. The normal adult heart rate is about:

(a) 40 beats per minute
(b) 70 beats per minute
(c) 100 beats per minute
(d) 120 beats per minute

74. The heart pumps approximately how much blood per minute at rest?

(a) 5 litres
(b) 10 litres
(c) 20 litres
(d) 1 litre

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ANSWER KEY WITH BRIEF REASONS

1. (b) Between the lungs in the chest cavity — It lies in the mediastinum.

2. (b) Pericardium — Pericardium protects the heart.

3. (d) Protects against shock and prevents over-dilation — It is strong connective tissue.

4. (a) Pericardial fluid — Fluid reduces friction.

5. (c) Myocardium — Cardiac muscle forms myocardium.

6. (b) Endocardium — Endocardium is endothelial lining.

7. (b) Four — Two atria and two ventricles.

8. (a) Tricuspid valve — Right AV valve has three cusps.

9. (b) Bicuspid/mitral valve — Left AV valve has two cusps.

10. (b) Pulmonary artery and aorta — They prevent backflow into ventricles.

11. (a) Pumps blood to the whole body — Systemic circulation requires higher pressure.

12. (c) Venae cavae — Superior/inferior vena cava return body blood.

13. (c) Pulmonary veins — Pulmonary veins return lung blood.

14. (b) Deoxygenated blood to the lungs — It leaves the right ventricle.

15. (d) Oxygenated blood to the body — It leaves the left ventricle.

16. (b) Diastole — Diastole is relaxation/filling.

17. (b) 0.1 second — Atrial contraction is brief.

18. (b) 0.3 second — Ventricular contraction is longer.

19. (c) Atrioventricular valves — AV valves close at ventricular systole start.

20. (d) Semilunar valves — Semilunar valves close at ventricular relaxation.

21. (c) Turbulent blood flow through defective valves — Valve problems create abnormal sounds.

22. (c) SA node — SA node initiates impulses.

38. (b) Aorta — Velocity falls as cross-sectional area increases.

39. (d) Capillaries — Large total cross-section slows flow.

40. (d) Blood flow into capillaries — They control capillary perfusion.

41. (d) Increases blood flow to tissues — Relaxation widens vessels.

42. (c) Decreases blood flow — Contraction narrows vessels.

43. (d) Lungs — It oxygenates blood.

44. (c) All body parts except lungs — It supplies tissues.

45. (d) Heart wall — Coronary vessels nourish myocardium.

46. (d) Liver — It delivers absorbed nutrients.

47. (d) Aorta — Pressure falls along the circulation.

48. (a) 120 mm Hg — It reflects ventricular contraction.

49. (b) 80 mm Hg — It reflects ventricular relaxation.

50. (c) Sphygmomanometer — It measures arterial pressure.

51. (a) Carotid arteries and aortic arch — They monitor blood pressure.

52. (b) Increased heart rate and vasoconstriction — They restore pressure.

53. (d) Increasing water retention and vasoconstriction — It conserves water and constricts vessels.

54. (c) Increasing salt and water excretion — It reduces blood volume.

55. (a) Fatty materials/cholesterol in arteries — Plaques narrow arteries.

56. (a) Hardening of arterial walls — Calcification stiffens arteries.

57. (c) A blood clot within a vessel — It can block blood flow.

58. (d) A dislodged thrombus that travels in blood — It can lodge elsewhere.

59. (c) Death of heart muscle due to blocked blood supply — It is a heart attack.

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23. (b) Right atrium — It lies in the upper right atrial wall.

24. (a) Interatrial septum — It lies in the lower interatrial septum.

25. (d) Atria to complete contraction before ventricles contract — It coordinates sequential pumping.

26. (b) Ventricles via Purkinje fibres — It spreads excitation through ventricles.

27. (a) SA node is weak or impulses are blocked — It restores rhythmic contraction.

28. (a) Electrical activity of the heart — ECG traces cardiac electrical potentials.

29. (b) Away from the heart — Arteries are efferent vessels.

30. (a) Toward the heart — Veins are afferent vessels.

31. (c) Tunica media — Muscular/elastic media withstands pressure.

32. (d) A single layer of endothelial cells — Thin walls allow exchange.

33. (d) Capillaries — Capillaries are exchange vessels.

34. (b) Interstitial fluid — It bathes cells.

35. (c) Prevent backflow of blood — Low-pressure flow needs valves.

36. (c) Pulmonary arteries — Pulmonary arteries carry deoxygenated blood.

37. (d) Pulmonary veins — Pulmonary veins carry oxygenated blood.

60. (b) Chest pain due to insufficient oxygen to heart muscle — It signals oxygen demand/supply mismatch.

61. (a) Uses a graft to bypass a blocked coronary artery — It restores myocardial blood flow.

62. (b) Dilate a narrowed artery — It opens blocked vessels.

63. (a) A mesh tube that keeps an artery open — It supports the vessel wall.

64. (b) 140/90 mm Hg — Chronic elevation is hypertension.

65. (c) Brain, heart and kidneys — It stresses multiple organs.

66. (a) Interstitial fluid — It maintains fluid balance.

67. (a) Lacteals — They absorb fats.

68. (c) Lymph — They contain immune cells.

69. (b) Blood — It removes aged RBCs/pathogens.

70. (a) Left subclavian vein — It returns lymph to venous blood.

71. (a) Right subclavian vein — It drains the right upper body.

72. (c) Skeletal muscle contractions and valves — Muscle pump moves lymph.

73. (b) 70 beats per minute — Resting rate is around 70/min.

74. (a) 5 litres — Cardiac output is about 5 L/min.

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